

Comments on Wilson's *Physics Avoidance* for APA "Author Meets Critics" Session

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January 6, 2020

When I was first introduced to Mark Wilson's work almost thirty years ago, I found it amongst the most interesting and original work I had seen in philosophy. Of course, I was young and naive then, but my initial impression has only been reinforced over time. Very little work spans so many issues — issues in philosophy of language, in scientific reasoning, in ontology, etc. — at such a sophisticated level. I find the book under discussion today, *Physics Avoidance: Essays in Conceptual Strategy* to be classic Wilson. The only work which is comparable in range of topics and outlook is Wilson's prior book *Wandering Significance*. *Physics Avoidance* is an excellent complement to and further elaboration of the ideas in his prior book but it also fully stands alone and contains significantly more detail about the way reasoning between scales works in physics and what it suggests about the organization of our scientific information. This is an unusually rich book that really ought to set the stage for the topics it covers as the philosophical conversation goes forward. Whether it will or not depends upon whether philosophers are willing to let go of the comforting simplicities that Wilson identifies.

One of the comforting simplicities and a main target of the book is "Theory T syndrome." Wilson plausibly locates the principle presumptions of this syndrome in a number of contemporary authors. The basic "Theory T" picture is as follows:

1. Theories, including theories of physics, are given by axioms.
2. The axioms are the laws of the theory.
3. The basic logic for reaching conclusions is deductive logic, not all of the resources of, what Wilson calls, "Greater Mathematics."
4. The interpretation of individual sentences of the theory does not depend upon other sentences, such things as what initial or boundary conditions one applies to the laws of the theory.

Philosopher's have used this tidy picture to give accounts of how to locate “perfectly natural properties” — just look at the properties that appear in the axioms while making allowances that one can logically combine them to form new properties — and ontological commitment — just look at where the existential quantifiers are to see what the theory says exists. The latter being (roughly) Quine's notion of ontological commitment.

I take it too that one of the presumptions of Theory T syndrome is that one gives (or, at least, can *in principle* give) an axiom system that has, in one sense or other, “complete coverage” of the physical world.¹ Presumably, the idea is that one axiomatizes “fundamental physics” and then one just reasons with it without any need for any other principles that, say, are descriptive of the macroscopic only. One simply reasons up from the bottom and only reasons in that direction. The bottom level (presumed to exist) completely covers all higher levels.

One of the novel features of Wilson's picture involves the way in which one sometimes reasons from “the top” (e.g. macroscopic) down — down to the mesoscopic, for instance — but, more than that, one sometimes reasons back up again. There is not simply a one-way transfer of information. One of Wilson's many examples is of modeling granite: One might start with a differential equation one initially takes to include a plausible description of the granite's macroscopic response; one solves that equation (under certain side conditions) while looking for signs that the pressure is enormous in some region; if one finds such a high pressure region, one takes that high pressure to be a sign that the granite has re-arranged itself at the mesoscopic scale in that region such that the original macroscopic equation is not suitable since the actual mesoscopic condition in that region does not give rise to the assumed macroscopic response; one then corrects the macroscopic equation in a way that takes into account the mesoscopic information. One might have to iterate this process until it settles on something definite.

Wilson notes how much in this way of reasoning goes against Theory T presumptions. Rather than deriving a differential equation from the “axioms,” one uses a predictor-corrector method to arrive at a better constitutive equation at the macroscopic scale via multi-scale modeling that looks into the mesoscopic scale only when needed. (There were hints of multi-scalar modeling in the earlier book but in this book it takes something closer to the center of the stage.) One does not even have an axiom system that tells one what constitutive equation to use. Even when attempts are made to axiomatize classical continuum mechanics (as in the Truesdell-Noll school),

¹I am not sure exactly how to make out this idea of complete coverage insofar as, I would expect, most Theory T-ists would think that physical theory as given in the axioms needs to be supplemented with initial and boundary conditions, even if such distinct types of conditions are blurred together as merely predicates in the antecedent of laws thought of as universal generalizations. So they do not think that everything true of the physical world follows from the axioms.

the “axioms” only give restrictions on constitutive equations; they do not say things like “If a material is granite, model it with this constitutive equation.” Moreover, all of this is taking place within “Greater Mathematics.” The reasoning is not simply deductive logic, let alone deductive logic from the “fundamental.” One of the grand themes of the book is the way we are able to exploit knowledge about macroscopic systems (or higher-scale systems in general) to model them without looking into the finer details or, at least, while looking into them as little as possible. Even if we had an axiomatic theory of the most fundamental level that we could trust, it would not necessarily be profitable to reason from it. Wilson describes “the overriding objective of all of these essays” as “to direct the reader’s attention to the importance of the varied strategic patches into which profitable reasoning frequently divides.” [Wilson, 2017, 243]

Once we take into account Greater Mathematics, Wilson notes that we can see that there are hidden properties not mentioned directly in even the final differential equation that we use. Such properties can only be reached by techniques like (generalized) Fourier decomposition, not via combination using basic logical vocabulary. Moreover, how we interpret differential equations depends upon the adjoined data. If we model a boundary with sharp corners, we will need to interpret the derivatives in the differential equation in a weak sense. So we are not able to give a once-and-for-all interpretation of the statements of the theory. Furthermore, multi-scale modeling renders it difficult to apply Quine’s criteria of ontological commitment. One might have a material that at a macroscopic scale is homogeneous and at a mesoscopic scale that sees finer-grained detail is not. One will, thus, have different differential equations for the different scales for the exact same material. But, as Wilson puts it, differential equations greedily reach down descriptively to arbitrarily small scales since their solutions take values (where they exist) at every point within a continuous space and time region. So if we amalgamate these equations into a single “theory” we will get a contradiction. In reality, we only allow the different scales to talk to each other via homogenization procedures, not via direct comparison of what the equations say.

I find so much of *Physics Avoidance* — not to mention Wilson’s earlier works — to supply more plausible and significantly more well-informed accounts of the topics it covers than what one normally finds that I do not know whether I am quite able to adopt the role of “critic” as opposed to someone who occasionally felt certain temptations to react in certain ways, though while being agnostic about whether there was much of anything to the reaction. Sometimes I didn’t even have the temptation in my own voice but, rather, had only suspicions as to how some of Wilson’s opponents, often opponents whose outlook I do not at all share, will react. Of the issues I bring up here, the last issue, having to do with the notion of law, is closest to my heart — as I am currently writing a book on how the notion of

law works in physics. In any case, I am mostly interested in giving Wilson a jumping-off point to share with us his elaborate views, not with any particularly “critique” that needs to be addressed.

1 Relation to Cartwright’s View

On the surface, there is a certain commonality between Wilson’s view of physics and large portions, at least, of Nancy Cartwright’s view such that one might be tempted to highlight the commonality between their views. Of course, Wilson does not accept Cartwrightian arguments that Newton’s Law of Gravitation is false for the reasons that she claims it is. But, there is much commonality even in the metaphors they use — both use the metaphor of a “patchwork” to describe the way physics covers the world, for instance. [Wilson, 2017, 185] Like Wilson, Cartwright will doubt that one can give an axiom system that has wide coverage rather than giving reasoning patterns that work in patches. And, I believe she will fully agree that “... [I]f science can only produce theories of a patchwork construction, then philosophical expectations that universe-wide ‘models’ and ‘possible worlds’ will be forthcoming from future science should be approached with greater caution.” [Wilson, 2017, 185]

It is clear that there is a difference in tone and emphasis. Wilson notes that her gloss on the issues they discuss is “diametrically opposed” to his. [Wilson, 2017, xv, footnote 5] Cartwright’s claim that physics lies makes it sound as if something scandalous is happening whereas Wilson takes it that it is quite reasonable to start with simple, tractable problems, even if they elide certain details. It is sensible, not scandalous, to do so. Moreover, Cartwright takes certain reasoning patterns in physics (taking limits in a certain order) to be “put up jobs” (as Wilson puts it) whereas Wilson invites us to look for the deeper rationales which have often been discovered by applied mathematicians. It strikes me, though, that Cartwright might adopt a “this could be the best we can do in terms of rationalizing this reasoning” stance of a sort that Wilson sometimes adopts. Any given example, she might be willing to let go of, should a deeper rationale for it be found, while remaining convinced that there is some example or other that will hold up to motivate her worry about how it all fits together. Only in the fullness of time might she be vindicated or refuted.

This led me to wonder what could be said by way of confrontation between them that could be more directly adjudicated now, without waiting to see whether all of the reasoning patterns that Cartwright is suspicious of will ever find their deeper rationale. I suspect that they differ over how widespread “idealization” is in physics and what counts as an idealization. I take it that Cartwright thinks that when we model a certain table as being of uniform mass density, we have invoked an idealization, a simplified description that is not accurate; it is strictly-speaking false because tables

consist of molecules (where the density is higher) and empty space (where it is lower). A contrary viewpoint is the idea that a claim like

(S1) The table has uniform mass density.

might be true at the macroscopic scale even if

(S2) The table does not have uniform mass density.

is true at some much smaller scale that can see the molecules that make up the table as well as the space between them. Rather, one speaks of truth in a scale dependent way; this idea renders (S1), not an idealized description at the macroscopic scale, but simply a true one. I hesitate to attribute to Wilson any attitude towards a particular example, but he certainly does in places within his work endorse ideas that sentences like (S1) can count as straightforwardly true, at least in a particular context. So, presumably, they are not inaccurate, idealized descriptions. (I do not, myself, understand the notion of idealization well enough to know whether it leaves room for idealized but true descriptions as some authors use it. I am assuming here that it doesn't.)

Similar issues will infect whether physics uses idealized laws, if the constitutive equations of continuum mechanics are laws. If one has a material that is heterogeneous at the mesoscopic scale — consisting of components that have different constants of elasticity, for example — one might still model its macroscopic behavior via a constitutive equation that has uniform elastic constants throughout. (This is Wilson's sort of example of a macroscopic equation that arises from a lower-scale differential equation via a homogenization procedure.) Wilson will think, I take it, that the resulting macro-scale constitutive equation is true of the material (or, at least, true within some small regime of strain). Cartwright, on the other hand, will take it to be an idealized constitutive equation that does not accurately represent the material response. I think she would also say that about the mesoscopic constitutive equations that one uses for the different component materials in the overall heterogeneous material as well, the thinking being that the whole format of continuum physics idealizes matter as continuous, when really it is not continuous in this way.

Assuming neither of them would object to how I have cashed out their view, I am not sure quite how to adjudicate it. I certainly feel the pull of the idea that (S1) might be the right thing to say in a context in which one is doing macroscopic reasoning. But I don't always have a handle on whether I think it is true or even a handle on what could help me decide whether it is true. I sometimes think that, supposing it is the best description of the table's density available at the macroscopic scale (e.g. it is ratified by measurements that probe small but macroscopic length scales and it gives rise to accurate macroscopic predictions using macroscopic physics), it is hard to see what would keep it from being true. Only a belief in the primacy

of the smaller – what Wilson calls “the Laplacean outlook” in some of his work — could make one want to balk at its truth.

As to the issue of whether classical continuum mechanics involves the idealization that matter is continuous, I don’t find myself tempted to say that the sentence

(S3) Matter is continuous.

is true at the macroscopic scale. Rather, I would be inclined to say that matter *seems* continuous at the macroscopic scale rather than *is* continuous at that scale. (This could just be the Laplacean outlook again but, if it is, I find it much easier to embrace it in this sort of example.) Hence, I don’t think that I would accept the reasoning that continuum mechanics is not engaged in an idealization when it models hunks of matter as continuous because the picture of matter it provides is accurate at that scale. This is a completely different matter from whether it is profitable to model it as continuous, which, as it happens, it is, but that is also a different matter from whether it is true that matter is continuous (even at that scale).

On the whole, though, I find myself rather at sea as to whether I should think that describing hunks of matter as possibly having uniform density, for example, involves an idealization. Maybe, though, so much the worse for the notion of idealization. I gather that Wilson also has this sense or, at least, might have this sense. Maybe he would be inclined to think that it doesn’t much matter whether any particular description is an idealized description or not since it doesn’t help us to be told that a certain description involves idealization. Whether it does or doesn’t involve idealization, one might still wonder why treating a table as a continuum macroscopically works as well as it does and why treating a table as having uniform density works as well as it might given the underlying molecular reality. Sometimes Wilson’s attitude seems to be along the lines of “sure there is lots of simplified description going on in physics”, but the claim that science idealizes should never be one’s conclusion. It should be the starting point of an investigation.² [Wilson, 2017, 140] To be told that homogenization methods result in idealized equations (even if it is true) and that so do quasi-static assumptions is not to learn much about how science proceeds since different cases of “idealization” are different and will be rationalized differently. Perhaps, what can sound like a grand revolutionary thesis is actually quite humdrum, not amounting to much more than that the world is quite complicated and so to get a reasoning foothold we often have to move to something tractable

²I believe that Cartwright would agree with this much. She attempts to rationalize some of the procedures that go under the rubric “idealization” via appeal to the capacities that objects have that are realized in simple situations but which carry over to more complex situations in which multiple capacities are at work. I am skeptical, as I would suspect Wilson is, of whether this story can be told in any illuminating way to rationalize the sorts of examples that Wilson explores. I am even skeptical of it in the cases that Cartwright discusses.

and try to convince ourselves that we are getting information about the more difficult case out of it. I believe that this captures Wilson’s attitude rather than his being inclined to deny that physics frequently involves idealization, but I would be interested in hearing more about that.

Back to the possible Cartwright comparison, what is clear is that Wilson is walking a line between pristine Theory T thinking and a sort of Cartwrightian view that holds that the physical world is so messy that we cannot figure out how it all fits together. Wilson proposes that the sub-models in a multi-scalar modeling be thought of like charts of a differentiable manifold stitched together into an atlas. [Wilson, 2017, 237] This might be how future science is structured, rather than via axioms that cover the whole world, so to speak, like Theory T pictures. The emphasis in Wilson, though, is on how the patches are stitched together and the variety of different ways that might happen whereas in Cartwright there seems to be more skepticism about whether they can be stitched together in any reasonable manner.

2 Scientific Realism

I bring all of this up, not mostly out of interest in Wilson’s relation to Cartwright, something that mostly comes up in footnotes in the book, but because all of this interacts with the question of scientific realism. Presumably, the issue of how much idealization there is in physics is relevant to how often we should take a realist attitude towards its descriptions. Wilson claims to be a scientific realist, but I was not entirely sure what scientific realism amounted to for him. Other authors have also wondered about the nature of Wilson’s realism and how “stout” it really is. [Ryckman, 2018] Wilson doesn’t anywhere start talking about structures and preservation of structure through theory change, nor about whether there are individuals or only structures, etc. So I don’t imagine that he is thinking in the terms of the structural realists of any variety. But I also do not believe that he would willingly endorse the claim from his advisor, Putnam, that our best sciences accurately describe the world (at least if they are suitably “mature”).³

Maybe part of the issue with discerning Wilson’s brand of realism is that one suspects that it is a much more modest thesis than is standardly (though not universally) attached to the words “scientific realism.” Often “realism” is a very grandiose thesis, not just the claim that there are some claims about unobservable entities that warrant belief, for instance, but that all of the claims of a mature science do. On the opposite side are statements to the effect that every claim in physics involves some sort of idealization and cannot be believed. Wilson is, presumably, somewhere in the middle here.

³In some places Wilson describes certain possibilities as not threatening to scientific realism. For example, Wilson claims that the issue over whether mathematical opportunism or mathematical optimism is true has little to do with realism. [Wilson, 2017, 233]

Still, I don't think I quite know where that is. For instance, should I be a realist about the differential equations of continuum mechanics? Wilson never talks that way – i.e. in terms of being a realist about a particular linguistic item — and perhaps he has good grounds not to talk this way because what the exact content of some items (including differential equations) is depends upon what the adjoined data is, as I have already mentioned. It isn't clear what it is to believe what a differential equation says as a purely syntactic item waiting for an interpretation. Even bracketing this and assuming we are in a context where the meaning of the differential operators is clearly the classical one, I am unsure what, of all the things a solution to a differential equation says, I am supposed to believe according to Wilson.

Insofar as their solutions have implications for nature at every scale, no matter how small, as Wilson very interestingly discusses, we should not naively believe everything they claim (or seem to claim? — I am not sure which is the right way to put it) since we know they are not getting small-scale features of nature right. Wilson can still claim, as he does, that they contain genuine physical information, often involving properties that span the whole system (e.g. Fourier modes) and that cannot, therefore, be thought to be part of nature in extremely smallish regions. I found that discussion very interesting — particularly the discussion of how we lose accurate information about those properties (or at least we lose our standard tools of accessing it) if we substitute the differential equation with something like a finite difference equation that does not descriptively reach down to all scales. [Wilson, 2017, 216] But maybe this sort of thing is all that Wilson means by “realism”: Out of the equation one gets some genuine physical information even if a lot of what the solution to the differential equation says cannot be naively trusted; rather, we might have to do significant investigation to know exactly what of what it says can be trusted (e.g. one might not know in advance whether it is accurate at the mesoscopic scale or only at the macroscopic scale?). It is a cautious realism that maintains we should not just naively trust what a differential equation is saying but which also maintains some confidence that with further investigation, we can know what of what it says we can trust and, given our present knowledge, we often have a sense of this already. Does this capture the “stoutness” of Wilson's realism?

3 In Principle Possibility?

On a rather different note, Wilson is fully aware that many will think that they can avoid every example he supplies in the book by claiming that he is writing about how science works *in practice* whereas *in principle* everything can be done from the ground up without what they take to be the mere reasoning expediences that Wilson discusses. In various places, Wilson argues that we know that the approaches we have now that proceed from

the bottom are less reliable than a multi-scalar modeling of the situation and that we do not know that there are better methods. The idea that there are is just an “aspirational hope.” (One of the funnier lines in the book is when Wilson declares “I sometimes wish that word processors might administer a mild electric shock to authors whenever they type the phrase ‘in principle.’” [Wilson, 2017, 21] I would hope for a somewhat better algorithm for distributing shocks insofar as this would shock someone like Wilson who is critical of the notion but typed the phrase several times. Maybe he would be willing to endure the shock as long as the supporters of the idea also face it, demonstrating just how much he does not like the notion.)

Although I am not one of these people – give me a realistic account of how reasoning in science actually works rather than an account of how it might at some point work any day — I suspect that those who have this bent will be unconvinced by some of Wilson’s examples. They might even think that they can see their way to the *in principle* possibility in certain cases. For example, in certain cases in which homogenization procedures are used to arrive at a reliable macroscopic equation from a mesoscopic equation, we change equations because the numerical solution of the mesoscopic equation would require a smaller grid size than our current computers can feasibly handle. It is not necessarily that we do not have a reliable numerical method that we could theoretically use to extract good information from the mesoscopic equation. In such a case, the fans of in principle possibility will claim that we can clearly see that there is only a practical limitation here. Remove our actual computing limitations and the problem is solved. We might never, in fact, remove those limitations. But we can see what gets in the way of such reasoning and it is not any matter of principle.

Even if they are right about that, it is just one example. It does not give one any sense of how they will reason away Wilson’s other examples as merely involving *in practice* limitations. It does, however, raise the question of what anyone means by *in principle* possibility. Presumably, it involves what can happen if one removes certain limitations that are, in fact, present. But what all limitations should be removed to arrive at what is in principle possible? If one removes too many limitations (e.g. removes our inability to specify the solution to a differential equation at a given point merely at a glance at the equation), it is not clear that anything like reasoning is going on at all. There is just directly intuiting the consequences of an equation without any intervening reasoning steps. This is nothing even remotely like how real science works.

Still, I imagine that many will be inclined to think that metaphysical truth need not track how inference in science works. Scientific inference has to operate within the confines of our computational abilities, but what is the case does not involve the limitations in our reasoning capacities. So, perhaps, in metaphysics one precisely wants to operate under a notion that idealizes such limitations away. I take it that Wilson thinks that this is a mistake,

even when doing metaphysics, not just philosophy of science. As Wilson puts it, earlier attitudes on the part of philosophers, which more closely resemble his, “strongly contrast with those that prevail within philosophical circles today, in which considerations of ‘metaphysics’ are pursued without concern for the issues of mathematical capacity and linguistic representation raised here.” [Wilson, 2017, 239] But I suspect that there is more to be said to convince metaphysicians that metaphysics ought to be pursued with these limitations in mind.

4 Skepticism about the Law Notion?

Many these days have taken a negative attitude towards the idea that there are laws of nature, often due to the disparate nature of putative exemplars. (See [Mumford, 2004], for example.) Although Wilson does not make such a point in order to claim that there are no laws, he endorses the overall idea that the set of things called ‘laws’ is not terribly unified: “In truth, the doctrines we call ‘laws’ in real life comprise a rag-tag collection of classifications tied together only through various forms of historical accident. I find it astonishing that this fuzzy notion is now regarded as a key ingredient within philosophy’s early a priori arsenal (replacing the ‘laws of causality’ of earlier times).” [Wilson, 2017, 263n34]

I agree with Wilson’s major point in the vicinity of this claim. The idea that we acquire some early law concept that has any alignment with what actually appears in physics strikes me as implausible, as does the idea that the law notion can do the sort of work that philosophers hope for, such as “grounding” counterfactuals. I am, however, less inclined to dismiss the law notion as fuzzy. Wilson is not completely dismissive either, he continues with “On the other hand, a narrower construal of ‘laws’ as representing the constitutive equation ingredients required to build up the hyperbolic operators of an evolutionary modeling strikes me as a fairly firm notion and close to the intuitive considerations to which philosophers often appeal.” [Wilson, 2017, 263] This construal seems to me, however, overly narrow insofar as it does not mention the “general principles” of continuum mechanics — for example, the balance of linear momentum and the balance of moment of momentum — as laws. In fact, in some ways, the general principles are better exemplars than constitutive equations since they are not specific to a type of material the way constitutive equations are. They are more universal, one might say, in the way one expects laws to be.

This, of course, raises the question of why include both the general principles and the constitutive equations in the set of laws. I would think that the unity here would be that they are all equations that are used in the derivation of differential equations of motion. This is not to say that everything that goes under the rubric ‘law’ has this status. Snell’s law and Kepler’s law of ellipses do not, though those two do not have the same sta-

tus. But many things that are laws have this status. (Even some of the claims of thermodynamics, such as the ideal gas law, that strictly only hold in equilibrium are used in the derivation of equations of motion for systems that are close to equilibrium via the assumption of local equilibrium.)

When I think of paradigm laws in this way, I am inclined to be more accepting even of those constitutive equations that lead to parabolic differential equations, such as Fourier’s heat law leading to the heat equation or the Navier-Stokes constitutive equation for the stress-tensor for a linearly viscous fluid. I know that one can protest that those cannot be real-world laws since, in the real world, there is not instantaneous transmission of signals across space as there would be if parabolic models were fully correct. This is, possibly, why Wilson talks of constitutive equations that lead to hyperbolic equations in the quote just given. But if one asks “What laws are we using in this model?”, Fourier’s heat law might be perfectly acceptable and true. I, myself, would not even balk at saying that it is both approximately a law of the world but also that it is a law of classical continuum mechanics (at least as normally formulated; some variants include that the derived equations must be hyperbolic and it is forbidden within those variants.).

I think that Wilson is mostly claiming that the way philosophers invoke the law notion is fuzzy. I definitely agree with him there. But, I am inclined to think that the notion as it is actually used in physics is not that fuzzy. For many examples of laws (perhaps all), it is true that we can often think of reasons for calling it a law and reasons against it. For example, constitutive equations are clearly not physically necessary and are quite defeasible but, at the same time, they play the exact same role in “closing” Cauchy’s first law as Newton’s law of gravity plays in closing Newton’s second law. But this is just to say (I would have thought) that there are competing “directivities” on the word ‘law’. But much of Wilson’s work has been focused on how common this is, even for scientific vocabulary like ‘force’, ‘temperature’, etc. It is probably true that the word ‘law’ does not have the scientific utility of these other examples, but I do not think that it is totally without utility. Some of its utility might be the sort of “language management” utility that Wilson argues a word like ‘cause’ displays. If we find ourselves with a bad model, we might wonder whether the problem is with one of the laws we used to derive the equation of motion or with something else. (Though here, we are likely to settle on the constitutive equation as the culprit, which could be why Wilson mentions them in what he says about laws.) I would not be surprised if Wilson were to agree with all of this. The nay-sayers about laws (for example van Fraassen) are claiming something significantly too strong, just like the deniers of force were.

Wilson brings up the question of whether concrete, evolutionary ordinary differential equations are laws since they have the content to descriptively push a system forward in time in a way that something like Newton’s Universal Gravitation does not, at least not all by itself. [Wilson, 2017, 339]

In my work, I have tended to dismiss the idea that concrete, evolutionary differential equations are laws and to attribute the belief that laws have that sort of content to a mistake. I don't know who the mistake is first due to, but I see it as originating in philosophy. I find it present in David Hume, for example.

Within physics, however, things seem somewhat different to me. One does not have to think that there are ultimately axioms of continuum mechanics that the field will settle upon to think that continuum mechanics is best organized by separating out the general principles and constitutive equations rather than giving equations — like differential equations — that are a blend of both. Insofar as considerations of derivational primacy supplies some pressure on the word 'law,' one would expect it to pull away from calling differential equations 'laws,' as it seems to me to do. Standard conventions for naming equations in physics seem to me to respect the distinction between laws and evolutionary differential equations. When differential equations of evolutionary type have names — like 'wave equation' or 'heat equation' — they tend not to involve the word 'law' whereas the constitutive equations ('Hooke's law' or 'Fourier heat law') and general principles ('Newton's second law' and 'Cauchy's First law') do.

This is not to say that one cannot find both physicists and philosophers referring to evolutionary differential equations as laws. One certainly can, but I see this as the prolongation of a mistake, one that we can recognize as a bad path to have gone down. I don't take this to be inconsistent with anything Wilson says since mistakes can certainly supply pressure on words. In the end, we both agree that if the law notion is to have non-confusing application various distinctions need to be drawn. Maybe the only place that we truly differ is that I am more optimistic that there is a unity to the law notion that encompasses most major examples. But I am not even sure that we differ there.

All of that said, where Wilson provides an extremely useful corrective to much discussion of the role that laws play in physics is in noting that 1) classical physics has never settled on a complete list of laws (for example, no laws for the force of cohesion have been provided [Wilson, 2017, 268]) and 2) we often engage in reasoning that has, as its main purpose, the *avoidance* of laws (if we can assume that a piece of material is rigid, we can ignore the complicated laws governing its interior). Theory T presumptions would have expected the contrary of both of these. Wilson is also undoubtedly right that the 'law' notion has been invoked inappropriately to provide overly tidy pictures of counterfactuals, natural properties, etc. But to me, these points indicate that we have a tolerable handle on what the law notion predominantly tracks in physics. We would not know that we are engaging in law avoidance if we did not know what laws are. For, we would not know what sort of thing we are avoiding.

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